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Real-Time Hydrocarbon Volume Fraction Prediction Using Machine Learning: A Sensitivity Analysis of Different Data Sources

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Abstract

Hydrocarbon volume fraction (HVF), the ratio of the hydrocarbon content of a rock to the total volume of all fluids and usually expressed as a fraction, is one of the essential reservoir properties, similar to water saturation, used for the reservoir volumetric assessment. The traditional methods of estimating HVF from core analysis and wireline log analysis have limitations. This study, through a robust sensitivity analysis, investigates the data source that best predicts HVF.

We investigated the individual predictive capability of drilling parameters, advanced mud gas measurements, and cuttings lithology to predict HVF by running three groups of experiments. Datasets from five wells were used to train three machine learning (ML) algorithms: artificial neural networks (ANN), decision tree (DT), and random forest (RF). The database is randomly split into training and testing subsets in the 70:30 ratio. We also implemented the multivariate linear regression (MLR) algorithm as a benchmark. Each group were tested for their respective capability to predict HVF.

The comparative results showed that the drilling parameters and mud gas data are capable of predicting HVF with the RF model emerging the best in both cases. The RF model recorded the highest training and testing R-Square values of 0.98 and 0.96 for the drilling parameters and 0.99 and 0.94 respectively for the mud gas data. The use of cuttings lithology to predict HVF was found to be unpromising with the ANN model recording the highest training and testing R-Square values of 0.30 and 0.27 respectively. Within the drilling parameters group, the MLR model gave the least training and testing R-Square values of 0.47 and 0.38 respectively in favor of the ML models with much higher values. This indicates that the relationship between the drilling parameters and HVF is nonlinear. Hence, HVF could not be accurately predicted using a linear model such as MLR. In the mud gas data group, all the models performed well, but the ML models did much better than the MLR. For the cuttings lithology group, all models performed poorly. This indicates that the ML methodology, along with other data-driven approaches, is still guided by the physical laws that are represented in the data.

This study has proven that using either drilling parameters or mud gas data only will give the same level of accuracy, or even better, in real-time HVF predicting, among other benefits. This partly addresses the

challenge of data availability and confirms that "less could be better". Our future work will investigate an optimal subset of features from all the real-time data that will produce the most accurate prediction of HVF.

Introduction

Hydrocarbon volume fraction (HVF) is the proportion of hydrocarbons (oil and gas) compared to the total volume of the entire fluids in a rock sample. It is a dimensionless quantity, usually expressed as a decimal, indicating the fraction of pore space within a rock occupied by hydrocarbons. It is an important input parameter for reservoir volumetric assessment. HVF is critical for the accurate estimation of the hydrocarbon volume in a reservoir, which is an important input for reservoir simulation, production forecasting, fluid contact determination, pay zone identification, and economic evaluation (Anifowose et al., 2024b). HVF is typically calculated using the formula: $HVF = \text{Pore Volume} * (1 - \text{Water Saturation})$, with porosity, net thickness, and water saturation as key inputs (Wheaton, 2016). These inputs are challenging to obtain, either from the analysis of wireline logs data or geological core analysis. Both cannot provide the data in real time.

Our previous studies have successfully proven the feasibility of predicting HVF in real time (Anifowose et al., 2024a, 2024b), utilizing mud gas data and leveraging the power of machine learning. Apart from the mud gas data obtained while drilling, there are other data that are available in real time. These comprise surface drilling parameters and cuttings lithology volumes. Mud gas data have been identified as a critical resource that could reveal a wealth of reservoir information (Haworth et al., 1985; Torlov et al., 2019; Anifowose and Mezghani, 2021; Anifowose et al., 2022). Surface drilling parameters have been confirmed to have strong correlation with fracture pressure, rock strength, and other reservoir properties (Al-Abduljabbar et al., 2020; Ahmed et al., 2021; Alfaraj et al., 2023). Lithology has been reported to have close links with hydrocarbon volume fraction (Su et al., 2022; Singh et al., 2023). Since there is no guarantee that all these data would be available at all times, it is important to determine the comparative relevance of each of them to the prediction of HVF. Hence, the main objective of this study is to investigate which of these data (mud gas, drilling parameters, and cuttings lithology volumes) will predict HVF most accurately.

Some of the benefits obtainable from the feasibility of predicting HVF in real time include (1) improved reservoir characterization, (2) enhanced drilling efficiency, (3) reduced drilling risk, and (4) increased well productivity. Real-time HVF estimation can provide more accurate and detailed understanding of reservoir properties while drilling, leading to better risk-mitigating decisions in real time. By providing early indications of productive zones, real-time HVF estimation can help to optimize drilling operations and reduce non-productive drilling time. Accurate HVF estimation can help prevent costly drilling complications such as kick and loss of circulation by identifying potential problems early on. Lastly, by targeting productive zones with more precision, real-time HVF estimation can lead to higher well productivity and increased oil and gas production.

Methodology

The methodology applied to this study followed the standard machine learning workflow comprising (1) identifying the three data sources, (2) extracting features from each data source, (3) combining extracted features for all available wells, (4) cleaning data, (5) analyze data correlation before and after cleaning, (6) building models for each data source, (7) select the best model, and (8) use the best model for blind validation. This workflow is depicted in Figure 1.

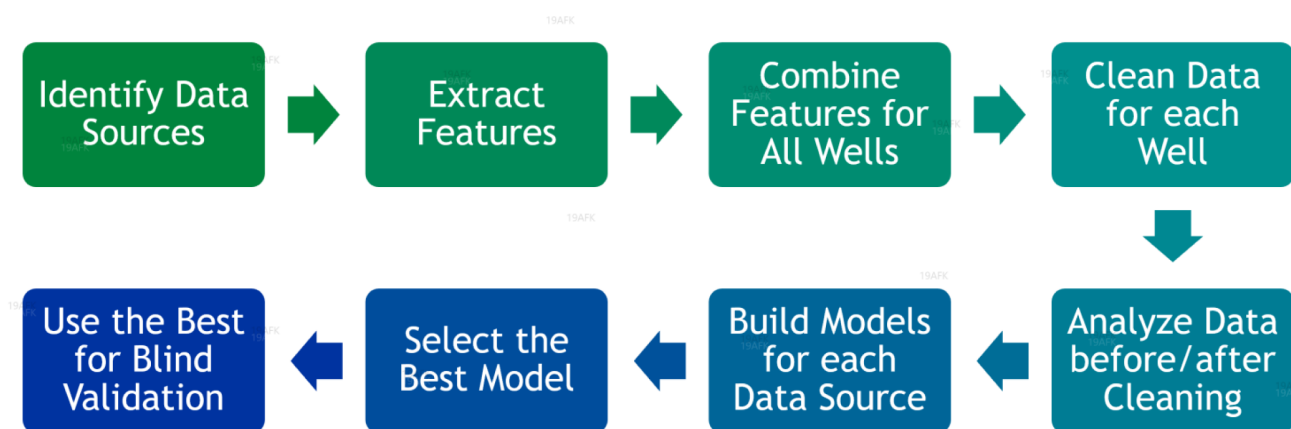


Figure 1—The machine learning workflow applied in this study.

Surface drilling parameters, mud gas, and cuttings lithology data were collected from seven wells. The data for five wells were combined to build a working database. This was then randomly split into training and testing subsets in the 70:30 ratio. Three ML algorithms namely feed forward neural networks (FForwardNN), regression tree (Reg Tree), and random forest (RanFor) were used to build models. A multivariate linear regression (LinR) model was added as a benchmark. The hyperparameters of all the models were tuned and optimized automatically using the Bayesian method. The remaining two wells were used for blind validation after selecting the best model. To select the best model, we used the two common performance evaluation metrics: correlation coefficient (R-Square) and mean absolute percentage error (MAPE).

The surface drilling parameters data have 17 features including bit size, hook load (HKLI), lag time, mud density (RHO_in and _out), rate of penetration (ROP), revolutions per minute (RPM), standpipe pressure (SPP), torque and weight on bit (WOB). The mud gas data have 16 features including light gases (C1 – C5), heavy gases (C6+), and the total gases (THC). The cuttings lithology volumes data have five features comprising volumes of anhydride, dolomite, limestone, sandstone, and shale. All the features from the three data sources are shown in Figure 2.

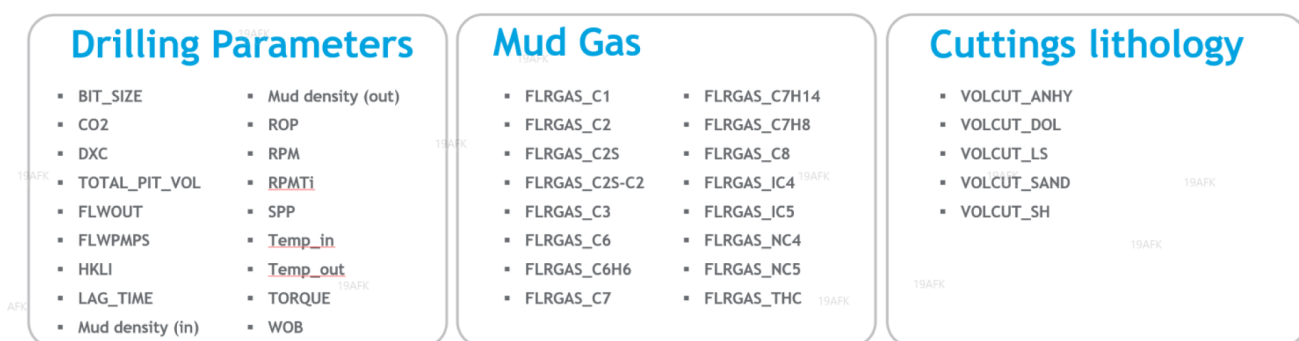


Figure 2—All the features from the three data sources.

To execute all the models, we used the codeless platform, AiLAB (Anifowose et al., 2025), developed inhouse and used by data scientists, ML engineers, and domain experts, including users with no coding skills. Using the AiLAB platform, we ran three experimental scenarios for each model comprising each of the three data sources.

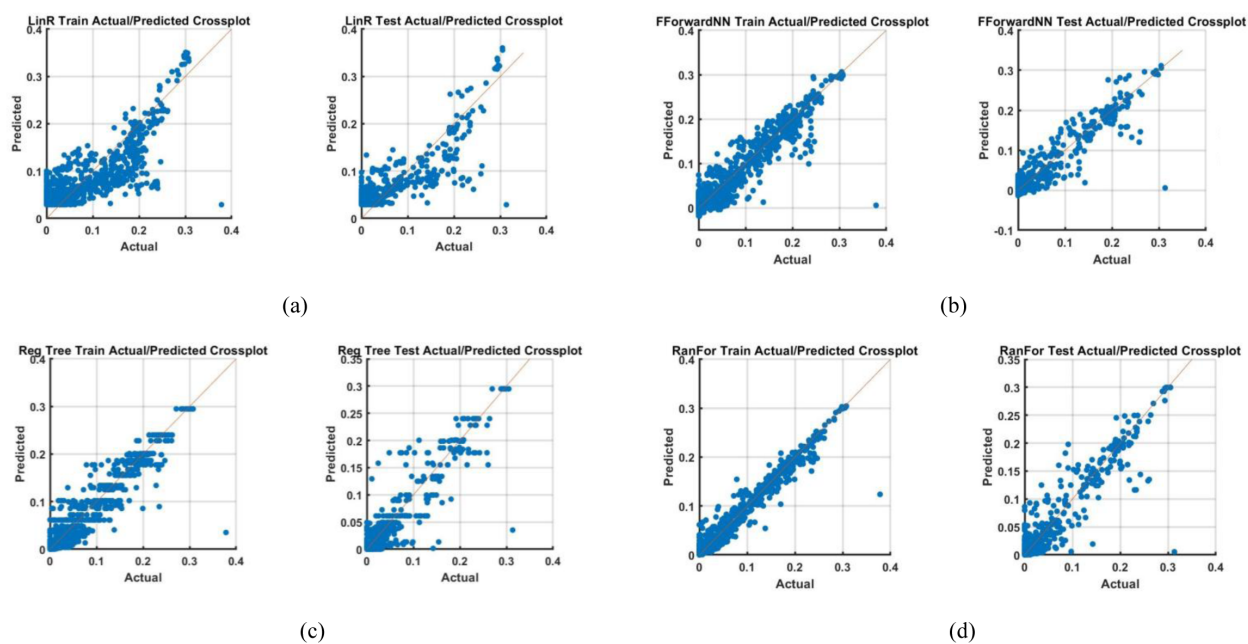
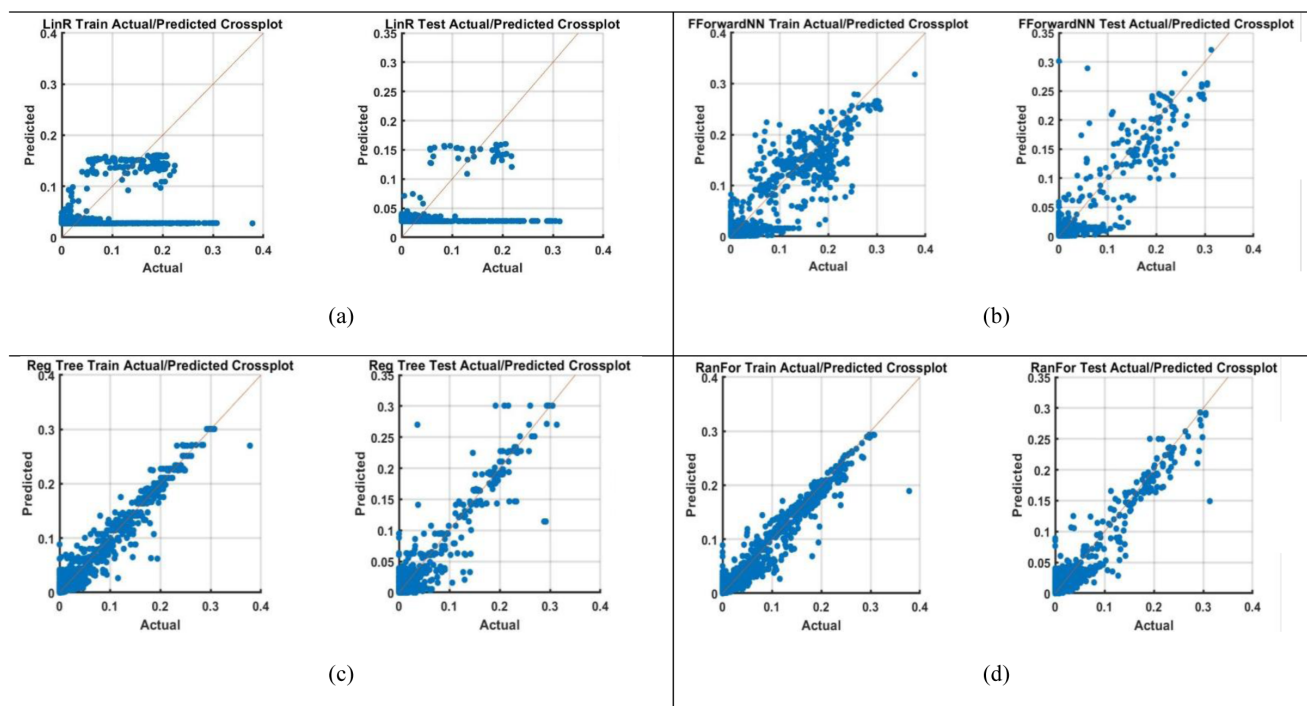
Results and Discussion

Starting with the dominant features recommended by each model with respect to each data source, the LinR and FForwardNN models suggested the scantiest number of features. For the surface drilling parameters, the LinR model recommended that only CO₂ (positive) concentration is the most linearly correlated to the HVF while FForwardNN recommended TEMP_MUD_OUT (negative), SPP (positive), RHO_MUD_IN (positive), HKLI (positive), FLWPMPS (negative), and BIT_SIZE (positive) as being among the most nonlinearly correlated. The Reg Tree and RanFor models have very close agreement that TEMP_MUD_OUT, SPP, RHO_MUD_OUT, and RHO_MUD_IN (all positive) are among the most nonlinearly dominant among the features. For the mud gas data, the LinR model presented only C1 and THC (both positive) as the most linearly correlated to HVF while FForwardNN recommended C1, C3, IC4 (all negative), and IC5 (positive) as the most nonlinearly correlated. Contrary to the surface drilling parameters, the Reg Tree model suggested all the features (all positive) to be relevant to the prediction of HVF in the nonlinear domain. According to the RanFor model, C1, C2, C2S, C3, C6, C7, IC4, IC5, and THC are the most dominant. For the cuttings lithology, three features (ANHY, DOL, and SAND) were excluded from the feature importance analysis as they are composed of zeros across all the five wells combined for this modeling task. Despite the indication that the cuttings lithology volumes are not good predictors of HVF, the LinR model showed that the SH is more linearly correlated (negative) than the LS (positive). The FForwardNN and Reg Tree models both agreed that the LS is more nonlinearly relevant than the SH (both positive). However, the RanFor model suggested that the SH is more dominant than the LS. The results are summarized in [Table 1](#).

Table 1—Summarized results of the feature importance analysis.

	Initial Number of Features	Dominant Features			
		LinR	FForwardNN	Reg Tree	RanFor
Drilling Parameters	17	One feature: CO ₂	Six features: TEMP_MUD_OUT, SPP, RHO_MUD_IN, HKLI, FLWPMPS, and BIT_SIZE	Four features: TEMP_MUD_OUT, SPP, RHO_MUD_OUT, and RHO_MUD_IN	Four features: TEMP_MUD_OUT, SPP, RHO_MUD_OUT, and RHO_MUD_IN
Mud Gas Data	16	Two features: C1 and THC	Four features: C1, C3, IC4, and IC5	All features	Nine features: C1, C2, C2S, C3, C6, C7, IC4, IC5, and THC
Cuttings Lithology Volumes	5	Two features: SH and LS	Two features: SH and LS	Two features: SH and LS	Two features: SH and LS

Since there are cases of attempting to divide by zero in all the models, the MAPE metric was not available for all the modeling experiments and scenarios. Hence, we based our performance evaluation of the models on only the R-Square, supported by visual comparisons. As shown in [Figure 3 – 5](#), the mud gas data demonstrated the best performance on all the models, though LinR has the lowest accuracy. This shows that the mud gas data possess the highest linear and nonlinear relevance to the prediction of HVF. The cuttings lithology volumes data performed the worst on all the models. This shows that there is no predictive correlation between them and HVF in both linear and nonlinear domains. For the surface drilling parameters, the LinR model performed badly, indicating that there is no linear correlation between them and HVF. However, the ML models performed better, indicating that there exists a good correlation in the nonlinear domain. The RanFor model showed to be the best on all the three experimental scenarios.



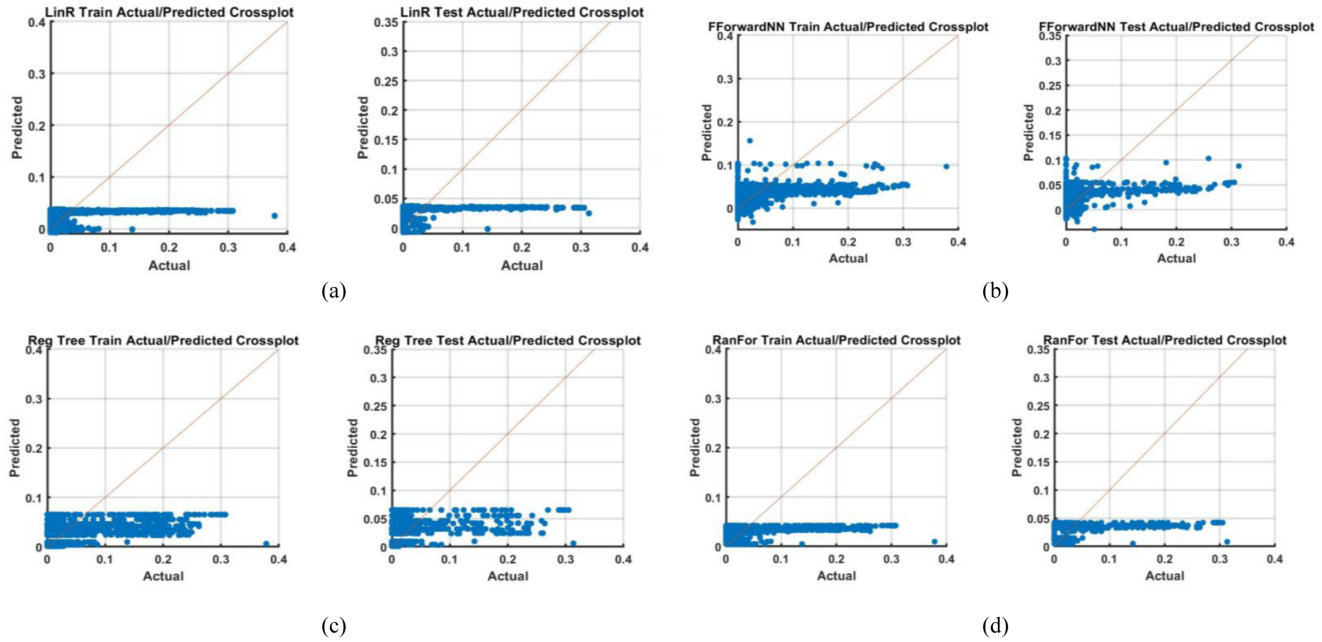


Figure 5—Comparative crossplots for (a) LinR, (b) FForwardNN, (c) Reg Tree, and (d) RanFor models for HVF prediction using cuttings lithology volumes.

Table 2 agrees with the comparative plots, showing that the drilling parameters and mud gas data are capable of predicting HVF with the RanFor model emerging the best in both cases. The RanFor model recorded the highest training and testing R-Square values of 0.98 and 0.96 for the drilling parameters and 0.99 and 0.94 respectively for the mud gas data. The use of cuttings lithology to predict HVF was found to be unpromising. This disagrees with what has been reported from [Su et al. \(2022\)](#) and [Singh et al. \(2023\)](#), calling for more work to validate this observation. For the surface drilling parameters, the LinR model gave the least training and testing R-Square values of 0.47 and 0.38 respectively in favor of the ML models with much higher values. This indicates that the relationship between the surface drilling parameters and HVF is nonlinear. Hence, HVF could not be accurately predicted from surface drilling parameters using a linear model. For the mud gas data, all the models performed well, but the ML models did much better than the LinR. This agrees perfectly with our hypothesis in earlier studies ([Anifowose et al., 2024a, 2024b](#)). The poor performance of all the models on the cuttings lithology volumes data indicates that the ML methodology, along with other data-driven approaches, is not a magic wand that must work in all situations, but strongly guided by the physical laws that are represented in the good quality data.

Table 2—Comparative results of all models over the three experimental data source scenarios.

	Model	Train RSquare	Val RSquare
Drilling Parameters	LinR	0.47	0.38
	FForwardNN	0.91	0.89
	Reg Tree	0.98	0.94
	RanFor	0.98	0.96
Mud Gas Data	LinR	0.86	0.86
	FForwardNN	0.95	0.94
	Reg Tree	0.96	0.94
	RanFor	0.99	0.94
Cuttings Lithology Volumes	LinR	0.16	0.15
	FForwardNN	0.3	0.27

Model	Train RSquare	Val RSquare
Reg Tree	0.33	0.25
RanFor	0.27	0.25

Conclusions

We presented in this study a comparative analysis of the predictability of HVF using three different sources of real-time data: surface drilling parameters, mud gas, and cuttings lithology volumes data. The objective was to determine which of these data sources will give the best prediction of HVF. This study builds on the previous ones that focused on the mud gas data, by extending the investigation to other sources of real-time data. Knowing this will help geoscientists and petroleum engineers to know which data source to focus on while attempting to predict HVF in real time. Utilizing a robust ML workflow involving four algorithms, we investigated the three scenarios, analyzed the results, and reported the following observations:

- The mud gas data emerged as the best to predict HVF, as they possess the highest linear and nonlinear relevance to the prediction of HVF.
- The cuttings lithology volumes data were found to have no predictive correlation with HVF in both linear and nonlinear domains.
- After the mud gas data, the surface drilling parameters have good performance with the ML models.
- All the tested scenarios confirmed that the ML models are more desirable to handle the nonlinearity in the datasets.
- Among the ML models, the RanFor model demonstrated the best performance on all the three scenarios.
- We confirmed that ML methodology is not a magic wand that must work in all situations, but strongly guided by the physical laws that are represented in the good quality data.

The outcome of this study contributes to the establishment of the feasibility of overcoming the limitations of the traditional methods of estimating HVF from core analysis and wireline log analysis. This study has proven that using mud gas data, or in the absence of that, surface drilling parameters, coupled with the ML methodology, will help to achieve the objective of predicting HVF in real time. This partly addresses the challenge of data availability and confirms that "less could be better".

Our future work will investigate the effect of combining all the data from the different sources on the predictability of HVF. Then, we will apply feature selection algorithms to extract an optimal subset of features from the combined data and also determine the effect of this.

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